Construction of Mimetic Curl Operators Using the MOLE Gradient

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1 Preliminaries

The Mimetic Operators Library Enhanced (MOLE) provides a one-dimensional mimetic gradient operator

$$\mathbf{G}_x = \texttt{grad}(k, m, \Delta x) \in \mathbb{R}^{(m+1)\times(m+2)},$$

where k is the order of accuracy and m is the number of cells. This operator maps scalar values at *cell centers plus boundary points* ($m + 2$ positions) to derivatives at *cell nodes* ($m + 1$ positions). The staggered positions are:

- **Nodes** (cell boundaries): $x_0, x_1, \ldots, x_m \quad (m + 1 \text{ points})$.
- **Centers + boundary**: $x_0,\ x_{1/2},\ x_{3/2},\ \ldots,\ x_{m-1/2},\ x_m \quad (m + 2 \text{ points})$.

All multi-dimensional arrays use **ndgrid ordering** (column-major with x varying fastest), consistent with MOLE's convention. The Kronecker product $\otimes$ is used to extend 1D operators to higher dimensions.

2 2D Curl Operator

2.1 Continuous definition

For a 2D vector field $\boldsymbol{F} = (F_x, F_y)$, the scalar curl is

$$(\nabla \times \boldsymbol{F})_z = \frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}.$$

2.2 Staggering

On a grid with m cells in x and n cells in y:

Component	Grid positions	Size
F_x (x-faces)	x: nodes $(m + 1)$, $\quad y$: centers+bnd $(n + 2)$	$(m + 1)(n + 2)$
F_y (y-faces)	x: centers+bnd $(m + 2)$, $\quad y$: nodes $(n + 1)$	$(m + 2)(n + 1)$
curl($\boldsymbol{F}$) (nodes)	x: nodes $(m + 1)$, $\quad y$: nodes $(n + 1)$	$(m + 1)(n + 1)$

The input vector is assembled as $\boldsymbol{F} = [F_x;\ F_y]$ with total length $(m+1)(n+2)+(m+2)(n+1)$.

2.3 Construction

Let $\mathbf{G}_x = \texttt{grad}(k, m, \Delta x) \in \mathbb{R}^{(m+1)\times(m+2)}$ and $\mathbf{G}_y = \texttt{grad}(k, n, \Delta y) \in \mathbb{R}^{(n+1)\times(n+2)}$. Define the identity matrices $\mathbf{I}_{m+1} \in \mathbb{R}^{(m+1)\times(m+1)}$ and $\mathbf{I}_{n+1} \in \mathbb{R}^{(n+1)\times(n+1)}$.

Term $\partial F_x/\partial y$. F_x lives on an $(m{+}1) \times (n{+}2)$ grid. Applying $\mathbf{G}_y$ along y while preserving the x dimension gives

$$\frac{\partial F_x}{\partial y} \longleftrightarrow (\mathbf{G}_y \otimes \mathbf{I}_{m+1})\, F_x, \qquad (\mathbf{G}_y \otimes \mathbf{I}_{m+1}) \in \mathbb{R}^{(n+1)(m+1)\times(n+2)(m+1)}.$$

Term $\partial F_y/\partial x$. F_y lives on an $(m{+}2) \times (n{+}1)$ grid. Applying $\mathbf{G}_x$ along x while preserving the y dimension gives

$$\frac{\partial F_y}{\partial x} \longleftrightarrow (\mathbf{I}_{n+1} \otimes \mathbf{G}_x)\, F_y, \qquad (\mathbf{I}_{n+1} \otimes \mathbf{G}_x) \in \mathbb{R}^{(n+1)(m+1)\times(n+1)(m+2)}.$$

Assembled operator. The 2D curl matrix acts on $\boldsymbol{F} = [F_x;\ F_y]$ as:

$$\boxed{\mathbf{C}_{2D} = \Big[\ -(\mathbf{G}_y \otimes \mathbf{I}_{m+1}) \quad (\mathbf{I}_{n+1} \otimes \mathbf{G}_x)\ \Big] \ \in\ \mathbb{R}^{(m+1)(n+1)\ \times\ [(m+1)(n+2)+(m+2)(n+1)]}}$$

so that $\mathbf{C}_{2D}\, \boldsymbol{F} = \partial F_y/\partial x - \partial F_x/\partial y$.

2.4 Curl of gradient identity

The full 2D gradient (without boundary injection) maps a scalar field ϕ at $(m{+}2)(n{+}2)$ cell-center positions to a face-staggered vector:

$$\mathbf{G}_{\text{full}} = \begin{bmatrix} \mathbf{I}_{n+2} \otimes \mathbf{G}_x \\ \mathbf{G}_y \otimes \mathbf{I}_{m+2} \end{bmatrix}.$$

Then

$$\begin{aligned}
\mathbf{C}_{2D}\, \mathbf{G}_{\text{full}} &= -(\mathbf{G}_y \otimes \mathbf{I}_{m+1})(\mathbf{I}_{n+2} \otimes \mathbf{G}_x) + (\mathbf{I}_{n+1} \otimes \mathbf{G}_x)(\mathbf{G}_y \otimes \mathbf{I}_{m+2}) \\
&= -(\mathbf{G}_y\mathbf{I}_{n+2}) \otimes (\mathbf{I}_{m+1}\mathbf{G}_x) + (\mathbf{I}_{n+1}\mathbf{G}_y) \otimes (\mathbf{G}_x\mathbf{I}_{m+2}) \\
&= -\mathbf{G}_y \otimes \mathbf{G}_x + \mathbf{G}_y \otimes \mathbf{G}_x \\
&= \mathbf{0},
\end{aligned}$$

using the mixed-product property $(\mathbf{A} \otimes \mathbf{B})(\mathbf{C} \otimes \mathbf{D}) = (\mathbf{AC}) \otimes (\mathbf{BD})$. This confirms $\nabla \times (\nabla\phi) = 0$ holds exactly at the discrete level.

3 3D Curl Operator

3.1 Continuous definition

For a 3D vector field $\boldsymbol{F} = (F_x, F_y, F_z)$:

$$\nabla \times \boldsymbol{F} = \begin{pmatrix} \partial F_z/\partial y - \partial F_y/\partial z \\ \partial F_x/\partial z - \partial F_z/\partial x \\ \partial F_y/\partial x - \partial F_x/\partial y \end{pmatrix}.$$

3.2 Staggering

On a grid with $m \times n \times o$ cells:

	Grid dimensions	Size
Input: face-staggered field		
F_x (x-faces)	$(m{+}1) \times (n{+}2) \times (o{+}2)$	$(m{+}1)(n{+}2)(o{+}2)$
F_y (y-faces)	$(m{+}2) \times (n{+}1) \times (o{+}2)$	$(m{+}2)(n{+}1)(o{+}2)$
F_z (z-faces)	$(m{+}2) \times (n{+}2) \times (o{+}1)$	$(m{+}2)(n{+}2)(o{+}1)$
Output: edge-staggered curl		
C_x (x-edges)	$(m{+}2) \times (n{+}1) \times (o{+}1)$	$(m{+}2)(n{+}1)(o{+}1)$
C_y (y-edges)	$(m{+}1) \times (n{+}2) \times (o{+}1)$	$(m{+}1)(n{+}2)(o{+}1)$
C_z (z-edges)	$(m{+}1) \times (n{+}1) \times (o{+}2)$	$(m{+}1)(n{+}1)(o{+}2)$

3.3 Construction

Let $\mathbf{G}_x$, $\mathbf{G}_y$, $\mathbf{G}_z$ be the 1D mimetic gradients of sizes $(m{+}1) \times (m{+}2)$, $(n{+}1) \times (n{+}2)$, and $(o{+}1) \times (o{+}2)$, respectively. Define identity matrices $\mathbf{I}_p$ of appropriate sizes with subscripts indicating dimension.

Each partial derivative is extended to 3D via Kronecker products. For a field on grid $(d_x \times d_y \times d_z)$, the operator applying $\mathbf{G}_\alpha$ along dimension α is:

$$\frac{\partial}{\partial z} \longleftrightarrow \mathbf{G}_z \otimes \mathbf{I}_{d_y} \otimes \mathbf{I}_{d_x}, \quad \frac{\partial}{\partial y} \longleftrightarrow \mathbf{I}_{d_z} \otimes \mathbf{G}_y \otimes \mathbf{I}_{d_x}, \quad \frac{\partial}{\partial x} \longleftrightarrow \mathbf{I}_{d_z} \otimes \mathbf{I}_{d_y} \otimes \mathbf{G}_x.$$

Applying this rule to each term of the curl, with the appropriate grid dimensions for each input component:

Term	Input grid	Kronecker form
$\partial F_z/\partial y$	$(m{+}2, n{+}2, o{+}1)$	$\mathbf{I}_{o+1} \otimes \mathbf{G}_y \otimes \mathbf{I}_{m+2}$
$\partial F_y/\partial z$	$(m{+}2, n{+}1, o{+}2)$	$\mathbf{G}_z \otimes \mathbf{I}_{n+1} \otimes \mathbf{I}_{m+2}$
$\partial F_x/\partial z$	$(m{+}1, n{+}2, o{+}2)$	$\mathbf{G}_z \otimes \mathbf{I}_{n+2} \otimes \mathbf{I}_{m+1}$
$\partial F_z/\partial x$	$(m{+}2, n{+}2, o{+}1)$	$\mathbf{I}_{o+1} \otimes \mathbf{I}_{n+2} \otimes \mathbf{G}_x$
$\partial F_y/\partial x$	$(m{+}2, n{+}1, o{+}2)$	$\mathbf{I}_{o+2} \otimes \mathbf{I}_{n+1} \otimes \mathbf{G}_x$
$\partial F_x/\partial y$	$(m{+}1, n{+}2, o{+}2)$	$\mathbf{I}_{o+2} \otimes \mathbf{G}_y \otimes \mathbf{I}_{m+1}$

3.4 Assembled operator

Let $s_{F_x} = (m{+}1)(n{+}2)(o{+}2)$, $s_{F_y} = (m{+}2)(n{+}1)(o{+}2)$, $s_{F_z} = (m{+}2)(n{+}2)(o{+}1)$, and similarly s_{C_x}, s_{C_y}, s_{C_z} for the output edge sizes. The 3×3 block curl matrix acting on $\boldsymbol{F} = [F_x;\ F_y;\ F_z]$ is:

$$\boxed{\mathbf{C}_{3D} = \begin{bmatrix} \mathbf{0}_{s_{C_x} \times s_{F_x}} & -(\mathbf{G}_z \otimes \mathbf{I}_{n+1} \otimes \mathbf{I}_{m+2}) & +(\mathbf{I}_{o+1} \otimes \mathbf{G}_y \otimes \mathbf{I}_{m+2}) \\ +(\mathbf{G}_z \otimes \mathbf{I}_{n+2} \otimes \mathbf{I}_{m+1}) & \mathbf{0}_{s_{C_y} \times s_{F_y}} & -(\mathbf{I}_{o+1} \otimes \mathbf{I}_{n+2} \otimes \mathbf{G}_x) \\ -(\mathbf{I}_{o+2} \otimes \mathbf{G}_y \otimes \mathbf{I}_{m+1}) & +(\mathbf{I}_{o+2} \otimes \mathbf{I}_{n+1} \otimes \mathbf{G}_x) & \mathbf{0}_{s_{C_z} \times s_{F_z}} \end{bmatrix}}$$

Note the antisymmetric sign pattern: the diagonal blocks are zero, and off-diagonal blocks appear with opposite signs across the diagonal, mirroring the structure of the continuous curl as a cross product.

3.5 Curl of gradient identity in 3D

The full 3D gradient maps a scalar field ϕ at $(m+2)(n+2)(o+2)$ positions to face-staggered components:

$$\mathbf{G}_{\text{full}} = \begin{bmatrix} \mathbf{I}_{o+2} \otimes \mathbf{I}_{n+2} \otimes \mathbf{G}_x \\ \mathbf{I}_{o+2} \otimes \mathbf{G}_y \otimes \mathbf{I}_{m+2} \\ \mathbf{G}_z \otimes \mathbf{I}_{n+2} \otimes \mathbf{I}_{m+2} \end{bmatrix}.$$

By the same mixed-product argument as in 2D, each row of $\mathbf{C}_{3D}\,\mathbf{G}_{\text{full}}$ reduces to the difference of two identical Kronecker products and therefore vanishes:

$$\mathbf{C}_{3D}\,\mathbf{G}_{\text{full}} = \mathbf{0}.$$

4 Summary

Both operators are constructed entirely from the 1D mimetic gradient provided by MOLE, extended to multiple dimensions through Kronecker products with identity matrices. The key properties are:

1. The operators are **highly sparse**.
2. The fundamental identity $\nabla \times (\nabla\phi) = 0$ **holds exactly** at the discrete level, by algebraic cancellation.
3. The operators inherit the **order of accuracy** k of the underlying mimetic gradient.